Designing The Concord: A Four-Domain Research Foundation

The Concord's most critical design challenge is not any single system — it is how economy, governance, AI threat, and long-arc progression interact to create a civilization that feels alive across decades of real time. This research synthesizes lessons from 20+ games, academic literature, and real-world governance theory into actionable design principles across four interconnected domains. The central finding: every successful persistent world derives its longevity from social bonds, not content — and every catastrophic failure traces back to either uncontrolled economics or insufficient player agency. The KALON contribution-based economy, the Assembly governance system, the Ascendancy AI, and the five-Phase arc must be designed as a single interlocking system, not four independent features.

DOMAIN 1: Virtual Economy Design — What History Teaches About KALON

EVE Online proved that economies need professional oversight and transparent destruction

EVE Online remains the most instructive virtual economy ever built. CCP Games made history in 2007 by hiring **Dr. Eyjólfur Guðmundsson**, CCP Games a PhD economist from the University of Akureyri, as the world's first in-house virtual world economist. Neoseeker His approach — modeled on being "a research scientist for a central bank" Fast Company — established a paradigm: publish transparent economic data, monitor inflation and deflation, and intervene only when systems are broken, never when players behave unexpectedly. Fast Company CCP's Monthly Economic Reports (MERs) track ISK money supply, velocity, price indices, sinks, and faucets with the rigor of a national statistics office. EVE Community As of late 2025, CCP hired **Stefán Þórarinsson** from the Central Bank of Iceland to continue this tradition for EVE Frontier. Blogger CCP Games

The ISK economy's defining insight is that **destruction is the primary economic regulator**. The economy follows a natural cycle: resources accumulate through ISK faucets (NPC bounties, mission rewards), geopolitical tension builds, and wars destroy vast quantities of assets. The Battle of B-R5RB in January 2014 destroyed **11 trillion ISK** (~$300,000 USD equivalent) in a single 21-hour fight. Wikipedia In November 2025 alone, roughly 730,000 hulls and structures were destroyed, The Ancient Gaming Noob with total losses hitting **74.85 trillion ISK** in that month. The Ancient Gaming Noob This destruction isn't a bug — it's the economy's circulatory system. Additional sinks include transaction taxes, broker fees, sovereignty payments (~1B ISK per system claim), and asset safety recovery fees, which hit 25.5 trillion ISK in December 2025 during the Pandemic Horde collapse. The Ancient Gaming Noob

The **PLEX system** bridges ISK and real currency without creating new money. Players buy PLEX from CCP with real money, then sell it to other players on the in-game market for ISK. Wikipedia Crucially, no ISK is spawned — it is pure player-to-player arbitrage. Ten Ton Hammer The reverse (selling ISK for real money) is prohibited, Wikipedia deliberately avoiding the regulatory exposure of a two-way exchange. CCP maintains reserve PLEX stocks to stabilize prices but intervenes cautiously. Fast Company The **Scarcity Era** (2020-2021) demonstrated CCP's willingness to impose macro-level resource restrictions, deliberately constraining supply to combat inflation.

The T20 scandal and Casino War reveal what happens when trust breaks

The **T20 scandal** (2007) was EVE's Watergate. A CCP developer operating under the alias "T20" spawned six rare Tech 2 Blueprint Originals and gave them to his corporation, Reikoku, Wikipedia part of the Band of Brothers alliance. INN Player Kugutsumen discovered the corruption by hacking an enemy forum. CCP apologized but did not fire T20 Wikipedia — he departed in late 2008. The Escapist Kugutsumen was permanently banned for revealing T20's real name. Wikipedia The crisis directly led to CCP establishing an Internal Affairs Division Wikipedia and creating the Council of Stellar Management as a democratic accountability mechanism. EVE Online

The **Casino War / World War Bee** (2016) demonstrated how unregulated financial instruments corrupt gameplay. The player-run gambling site IWantISK amassed vast wealth — the top five users won over **87 trillion ISK** between them — and used it to bankroll a massive military coalition against the Imperium. CCP eventually banned all player-run gambling services Massively Overpowered in late 2016. The lesson: any system where financial power translates directly to military or political power will be exploited at scale.

Second Life's Ginko Financial collapse and the limits of libertarian economics

Second Life's Linden Dollar economy peaked at a **$567 million GDP** in 2009, Secondlife24h with users cashing out roughly **$60 million annually** Wikipedia through the LindeX exchange at a stable rate of L$240-270 per $1 USD. Wikipedia But the economy's laissez-faire design proved disastrously vulnerable to financial fraud.

Ginko Financial (2007) was the defining crisis. This in-world "bank" promised **69.7% annual interest** on Linden Dollar deposits, with investments likely flowing into in-game casinos. When Linden Lab banned gambling in July 2007, a bank run ensued Wikipedia — withdrawal requests hit L$55 million within hours against deposits totaling L$180 million. Wikipedia Remaining accounts (200 million Linden Dollars, Wikipedia roughly **$750,000 USD**) were forcibly converted to perpetual bonds. CFO.com Individual players lost up to $10,000 of real money. CFO.com Linden Lab subsequently banned all in-world banks that lacked real-world banking licenses. MIT Technology Review

Cornell economist Robert Bloomfield called the collapse "a lesson in the limits of libertarianism." NBC News Raph Koster, co-creator of Ultima Online, delivered the epitaph: **"The libertarian economic ideal does not survive the management of a virtual universe."** New World Notes This directly informs KALON's design — even a post-scarcity contribution-based system needs regulatory mechanisms.

Entropia Universe has sustained a real-money economy for 20+ years through controlled decay

Entropia Universe is the only MMO operating a **fixed real-money exchange rate** (10 PED = $1 USD) EntropiaPlayer with actual cash withdrawals (minimum $100, processed in up to 50 business days). EntropiaPlayer Launched publicly in 2003 by MindArk PE AB in Sweden, it has sustained this economy for over two decades. Mindark The record transactions are staggering: Club NEVERDIE sold for **$635,000 USD** in 2010, Crystal Palace Space Station auctioned for $330,000 in 2009, and in 2021, an 800,000-deed land offering sold out **$800,000 in 22 minutes**. Cision

MindArk controls inflation through **universal item decay**. All weapons, tools, and equipment degrade with use, creating constant reinvestment pressure. Loot return rates, crafting failure rates, and resource spawning are all centrally tuned. Inflation manifests as changes in item markup (items trading above PED value) rather than currency devaluation. The system works because MindArk retains total control — per the EULA, players technically own nothing. EntropiaPlanets This creates an uncomfortable paradox: real-money investment in assets the developer can unilaterally adjust.

New World's deflation death spiral shows what happens without economic architecture

Amazon's New World launched in September 2021 ScreenRant to **913,000+ concurrent players** and saw its economy collapse within weeks. ScreenRant The failure was architectural, not incidental. The game had excessive gold sinks (crafting costs, repair fees, housing taxes, listing fees, transaction taxes) paired with insufficient gold faucets (no NPC vendors for selling items, limited endgame gold generation). MMORPG.com After players exhausted quest-based income at level cap, more gold left circulation than entered it. TheGamer

Multiple **gold duplication exploits** compounded the crisis. A server transfer bug allowed players to send gold to friends, receive an error, log back in with gold restored while friends kept the duplicate. MMORPG.com A second exploit during the emergency transaction lock allowed city owners to duplicate gold through upgrade attempts. GamePressure Amazon suspended all wealth transfers — trading, guild treasuries, auctions, and direct trades — freezing the entire economy. GamePressure The deflation spiral was textbook: prices crashed to "mere pennies," TheGamer crafting became unprofitable (a rare axe costing 30-75 gold to craft required 40 gold to repair), and **players reverted to a barter system**. GamePressure

The root failure: no NPC vendor system as a price floor, transaction validation that trusted client-side data, and an economy designed around scarcity without ensuring sufficient currency generation. **KALON must ensure contribution rewards (faucets) always exceed contribution costs (sinks).**

Albion Online's Black Market is the most elegant economic innovation in recent MMO history

Albion Online, launched in 2017 by Sandbox Interactive, Wikipedia operates a **100% player-crafted economy** with full-loot PvP in dangerous zones. Every item in the game — including those dropped by monsters — is player-made. The innovation that makes this work is the **Black Market NPC system**, introduced with the Hector update.

The mechanism is simple and brilliant: when a mob is killed, the game checks the Black Market's inventory. If the item is in stock, the mob drops it and stock decreases by one. If not, a buy order is created. MMO-Champion Crafters sell player-made items to the Black Market NPC, which is funded by taking 20% of silver from mob kills (players receive the remaining 80%). MMO-Champion A "corruption" mechanic permanently destroys a percentage of purchased items, acting as an item sink. Albion Online Wiki Game Director Robin Henkys noted: **"As far as we are aware, no other game has done something like this before."** Albion Online

The system solved a fundamental tension: PvE content needs satisfying loot drops, but player-crafted economies are undermined by NPC-generated items. The Black Market makes both work simultaneously. Albion Online For KALON, this pattern suggests that a contribution-based economy needs **mediating systems that translate between different types of economic activity** without the developers manually balancing every interaction.

Star Citizen's $929 million warning about pre-launch economic promises

Star Citizen has raised approximately **$929 million** as of late 2025 Simulation Daily (projected to cross $1 billion by mid-2026), Outlook Respawn with 2025 alone generating $155 million — a 33% year-over-year increase and the project's biggest year ever. Simulation Daily The game remains in alpha. Wikipedia Ships sold for real money range from $45 to several thousand dollars, creating enormous player expectations about economic value. The planned **StarSim** (formerly Quantum) economy simulation — using virtual AI agents called "Quanta" to simulate billions of NPCs making economic decisions Star Citizen Wiki — remains undelivered.

The lesson is sobering: economic promises made before a system exists create entitlement dynamics that constrain future design. Players who have spent thousands of dollars expect their ships to hold proportionate in-game value. **The Concord should never promise specific economic values or exchange rates before KALON is live and tested.**

The academic consensus: virtual economies mirror real ones, faster and harder

Edward Castronova (Indiana University) pioneered virtual economy research with his 2001

study of EverQuest's Norrath, Medium finding it had the **77th highest GNP per capita in the world** — between Russia and Bulgaria. Today I Found Out His key observation: even without subsistence needs, inequality rapidly emerges. Wealthy players hoarded resources and hired poorer players for menial tasks, reproducing real-world economic stratification Today I Found Out in weeks rather than centuries. His co-authored textbook *Virtual Economies: Design and Analysis* (MIT Press, 2014, with Vili Lehdonvirta of Oxford) remains the definitive reference. MIT Press

Yanis Varoufakis, later Greece's Finance Minister, was hired by Valve in 2012 to study Steam's virtual economy. Wikipedia Gabe Newell's reasoning was perfect: Varoufakis was an expert on the eurozone crisis, and Valve needed someone who understood exchange rate imbalances between connected economies — exactly the problem of linking TF2's hat economy to Dota 2's item economy. Yanis Varoufakis Varoufakis concluded: **"Economic theory has come to a dead end. The future is experimentation and simulation — and video game communities give us a chance to do all that."** Yanis Varoufakis

Dr. Marianne Johnson (University of Wisconsin) validated EVE's commodity markets as meeting all four conditions of a perfectly competitive market. Unswecosoc This is remarkable — virtual economies achieve theoretical economic ideals that real-world markets never fully realize, precisely because information flows more freely and transaction costs are lower.

Lessons for The Concord's KALON system

KALON's contribution-based design must solve five problems that destroyed other virtual economies:

First, contribution must have meaningful sinks. Without destruction or decay, contribution scores will inflate endlessly. The Ascendancy serves as KALON's equivalent of EVE's destruction mechanic — worlds lost to the Ascendancy should represent genuine economic loss that keeps the system in equilibrium.

Second, transparent economic data builds trust. Following CCP's MER model, publish regular KALON metrics — total contribution generated, contribution destroyed, velocity, distribution curves. Players who understand the economy invest in it rather than exploiting it.

Third, regulate financial instruments early. Second Life and EVE both learned that unregulated financial services (banking, gambling) attract exploitation. If KALON enables lending, insurance, or investment vehicles, build governance mechanisms before they're needed.

Fourth, design faucets before sinks. New World died because sinks exceeded faucets. Ensure that every way players can lose KALON is balanced by accessible ways to earn it, with a slight surplus to prevent deflation.

Fifth, consider intermediary systems like Albion's Black Market. KALON needs mechanisms that translate between different contribution types (military, economic, scientific, cultural) without making them directly fungible. A player defending a world from the Ascendancy and a player building infrastructure should both earn meaningful KALON, but the translation between these contributions needs a mediating system to prevent exploitation.

DOMAIN 2: Governance and Political Simulation — Building the Assembly

EVE's Council of Stellar Management is the longest-running democratic experiment in gaming

Created in 2008 in direct response to the T20 corruption scandal, (Hacker News) the CSM (EVE Online) is now in its 20th iteration. The council seats **10 elected members** plus 2 CCP-appointed members (introduced in CSM 18 to address representation gaps). Elections use **Single Transferable Vote (STV)** since CSM 8 (2013), where voters rank candidates and the lowest performer is eliminated each round with their votes redistributed. (Wikipedia) CCP publishes the full ballot data and Python STV script publicly — a remarkable commitment to electoral transparency. (EVE Online)

Voter turnout has fluctuated significantly: CSM 7 holds the all-time record at approximately 59,000+ votes, CSM 18 (2023) saw 47,155 votes, (Blogger) and CSM 20 (2025) reached **49,870 voters** across 55 candidates — the second-highest turnout ever. (EVE Online) The CSM has demonstrably shaped game development: members contributed feedback that influenced the PLEX system before launch, pushed for the widely popular Skill Training Queue, (EVE Online) and during the 2011 "Summer of Rage" (Monocle-gate), the CSM was called to Iceland for an emergency summit that resolved the crisis through direct negotiation with CCP leadership. (PC Gamer)

The most damaging criticism is **nullsec bloc capture**. Large player alliances (Goonswarm/Imperium, Pandemic Horde, Fraternity) consistently win 6-8 of 10 elected seats because they have organizational infrastructure for voter mobilization. (EVE News24) As one analyst noted: "To live in nullsec you need organization, and once you are organized you can use that for things like the election." (The Ancient Gaming Noob) Wormhole, lowsec, highsec, and small-gang PvP players feel chronically underrepresented. CCP's addition of 2 appointed seats was a partial corrective. (EVE Online)

The CSM has also been scandal-prone. The **Brisc Rubal Affair** (2019) damaged CCP's credibility severely: CSM 13 member Brisc Rubal was accused of leaking NDA information (EVE News24) for insider trading, permanently banned, and publicly denounced. **CCP later fully retracted all accusations**, issued formal apologies, and admitted their "initial actions were based on unsubstantiated assumptions." (Massively Overpowered) In 2024, CSM 18 member

Alcoholic Satan voluntarily admitted to being the source of an NDA leak about the Revenant expansion. EVE Online

The blue donut problem demonstrates how governance mechanics shape political behavior

EVE's sovereignty system has evolved through three major iterations, each attempting to prevent stagnation. **Dominion Sovereignty** (2009-2015) required destroying structures with tens of millions of hitpoints across multiple reinforcement timers, Eve-guides making conquest so tedious that alliances created non-aggression pacts rather than fight — the "blue donut" where mega-coalitions carve up all of nullsec. **Aegis Sovereignty / FozzieSov** (2015) replaced structure grinding with Entosis Link capture mechanics Eve-guides tied to system usage indices, but coalitions adapted rather than fracturing. The Ancient Gaming Noob **Equinox Sovereignty** (2024) introduced resource dependencies — power from stars, workforce from planets via Orbital Skyhooks, reagents for fuel — tying sovereignty to planetary exploitation and creating new raid targets. EVE Online

The Chinese Serenity server serves as the cautionary extreme: **one alliance achieved complete control**, leading to total stagnation. INN The blue donut is a Nash equilibrium — peace is individually rational because war is expensive and safe economic activity is profitable. EVE Online Breaking this equilibrium requires sovereignty mechanics that make peace costly or conflict rewarding, rather than merely making war easier. For The Concord, this suggests that **the Ascendancy should make complacent alliances more vulnerable than conflicting ones** — an external threat that punishes the blue donut.

Foxhole proves emergent leadership works for tactics but not strategy

Foxhole's two-faction system (Colonials vs. Wardens) operates with **no formal governance structures whatsoever**. Foxhole Wiki Leadership emerges purely through social dynamics: players gain influence through consistent contribution, expertise, and charisma. Regiments (the clan system) can hold up to 2,000 members Grokipedia and often mirror military structures with departments for logistics, combat, engineering, and HR. Grokipedia Wars last 20-30 days on average, Steam Community with victory requiring control of roughly 32 of 37 hex regions.

The system works for tactical coordination — moving supplies, organizing frontline pushes, conducting intelligence — because these tasks have clear feedback loops and short time horizons. But Foxhole lacks mechanisms for strategic-level governance: no way to set faction-wide priorities, resolve disputes between regiments, or make binding collective decisions. The result is that the most active and vocal players effectively control faction strategy through sheer presence, which creates exactly the "no-life capture" dynamic that The Concord's Assembly must avoid.

Crowfall's failure validated the concept while condemning the execution

Crowfall's vision of seasonal campaign worlds with permanent Eternal Kingdoms was arguably the most innovative governance design in MMO history. Campaign worlds progressed through Spring, Summer, Fall, and Winter seasons as "The Hunger" consumed the map, forcing players into smaller conflict zones. Flavor365 Results carried back to permanent Eternal Kingdoms where players could build, craft, and socialize between campaigns. Flavor365

The game launched in July 2021, Grokipedia peaked at roughly 3,000 concurrent players, MMOByte and shut down in November 2022. MMO Fallout Flavor365 The failures were execution, not concept: poor performance during large-scale sieges (the game's centerpiece), repetitive PvE, Flavor365 an eight-hour tutorial, Grokipedia and insufficient population for meaningful PvP. Most critically, **Eternal Kingdoms became ghost towns** — disconnected from the competitive experience, they had no reason to visit. The lesson for The Concord is that governance structures must have **mechanical consequences connected to core gameplay**, not exist in a disconnected social space.

Real-world governance models offer The Concord specific anti-capture mechanisms

Three real-world governance innovations are directly applicable to The Concord's Assembly:

Liquid democracy allows voters to either vote directly or delegate their vote to a proxy on any issue, with delegation revocable at any time. Wikipedia This solves the casual-player problem: players who can't attend every Assembly session delegate to trusted representatives but override on issues they care about. The Architect's 14% voting weight could function as a permanent delegate representing the interests of players who didn't vote.

Quadratic voting (conceived by economist E. Glen Weyl) gives voters "voice credits" where the cost of additional votes is the square of votes purchased SSRN — 1 vote costs 1 credit, 2 votes cost 4, 3 votes cost 9. This lets minorities with intense preferences express that intensity while preventing any single faction from dominating. Legal scholar Eric Posner called it "the most important idea for law and public policy that has emerged from economics in the last ten years." Steve Omohundro No game has implemented it yet — The Concord would be first.

Sortition — selection by lottery, the primary governance mechanism of ancient Athens Wikipedia — ensures no faction can game representation. Short terms with rotation distribute participation, Participedia and random selection from the active player base (stratified by playstyle, region, or Phase of entry) prevents organized bloc capture. Academic research on combining these approaches is accelerating: a 2025 paper by Kovalchuk et al. presents the first universally composable protocol combining quadratic voting with liquid democracy, directly applicable to virtual governance. IACR

Lessons for The Concord's Assembly

The Founding Assembly vote should use quadratic voting. Give all players equal voice

credits earned through gameplay contribution (not time played). This ensures passionate minorities can push back against apathetic majorities while preventing whales or no-lifers from dominating.

Routine Assembly decisions should use liquid democracy with delegation. Casual players delegate to trusted representatives but can override on issues they care about. This maintains participation rates without requiring constant attendance.

Some Assembly seats should be filled by sortition. Random selection from active players, stratified by activity type and Phase of entry, ensures that organized blocs cannot monopolize representation. Term limits (1-2 months) with rotation prevent entrenchment — EVE learned this lesson and introduced three-year CSM term limits at CSM 15. EVE Online

The Architect's 14% weight is a novel and defensible mechanic. Position it explicitly as representing "the interests of players who didn't vote" — the silent majority — and as an institutional check against organized minority capture. Make its voting rationale transparent and explainable.

Governance must have mechanical consequences. Learn from Crowfall's ghost-town Eternal Kingdoms: Assembly decisions must affect resource distribution, Ascendancy response, territorial priorities, and Phase progression. Players will engage with governance only if it changes the game they play.

Design sovereignty mechanics that punish the blue donut. EVE's entire sovereignty history demonstrates that players will default to peace if peace is profitable. The Ancient Gaming Noob The Ascendancy should be mechanically more threatening to large, passive alliances than to smaller, active ones — making political consolidation strategically risky.

DOMAIN 3: AI Director Systems — Making the Ascendancy Feel Alive

Left 4 Dead's Director proves that naming your AI is half the battle

Valve's AI Director, modl.ai presented by Michael Booth at GDC 2009, Drexel University is fundamentally a **finite state machine** modl.ai managing pacing through a stress/relaxation cycle — not the sophisticated intelligence players imagine. It operates through three states (Build-Up, Sustain Peak, Relax), measuring stress per player based on threat proximity, number of attackers, and damage taken. modl.ai It spawns mobs at randomized intervals of **90-180 seconds**, akamaihd controls item placement by selecting from pre-placed spawn points, and targets the least-stressed player for new attacks. In Left 4 Dead 2, it can dynamically alter map geometry from pre-authored variation sets. Aigamedev

The system was refined through **three years of daily playtesting** (2005-2008), Drexel University and the result feels organic because the superposition of multiple population functions

(wanderers, mobs, specials, bosses, items) creates emergent variety. (akamaihd) But co-writer Chet Faliszek later admitted the truth: **"The most genius development twist was naming it 'The Director' and allowing fans to ascribe a personality to it that never existed. It was a random number generator that kind of just worked."** (Game Developer) This is a profound design lesson. Players will attribute intelligence, personality, and even malice to any system given a name and narrative context. The Ascendancy benefits enormously from simply being named and characterized, even before the AI itself is sophisticated.

Alien: Isolation's two-brain architecture is the template for intelligent threat design

Creative Assembly's Xenomorph AI uses a **dual-brain architecture** that separates omniscience from behavior. (WhatCulture.com) The **Director AI (macro brain)** always knows the player's exact location (medium) and manages pacing through a "Menace Gauge" (Medium) — when tension gets too high, it sends the Xenomorph into vents for respite. (Game Developer) It posts job requests to a Blackboard System ({GoalType: Search, Location: Medbay_Lab_A, Priority: 0.75, Duration: 90s}) (Medium) and periodically directs the Xenomorph toward the player's general area. (Game Developer) (medium)

The **Xenomorph AI (micro brain)** does *not* know the player's true location and must hunt using its own senses (TheGamer) (medium) — four view cones, hearing (sound events with location, volume, and type), (Game Developer) and "motion sight" for weak rear detection. (TheGamer) It runs on a massive hierarchical Behavior Tree (Medium) and must physically traverse the station with no teleporting. (TheGamer) The "learning" effect is actually a **gated behavior system**: the Director tracks player metrics (flamethrower uses, locker hides, table hides), and when a threshold is crossed, it unlocks dormant behavior tree branches (Medium) — aggressive locker-search routines, flamethrower flanking, under-table checking. **Player actions give the AI "permission" to use tools it already had**, creating the illusion of adaptation. (medium)

This architecture — an omniscient director constraining a "blind" hunter — maps directly to the Ascendancy's needs. The Strategic Mind knows everything; the forces players encounter on individual worlds must hunt and fight with realistic information constraints.

The Nemesis System patent blocks direct imitation until 2036

Warner Bros.' **US Patent No. 10,926,179** on the Nemesis System (from Shadow of Mordor/War) was filed in March 2015 and granted February 23, 2021. (Patentarcade) It broadly covers controlling game events where predefined interactions between a player and one NPC result in changing parameters of a second NPC. (Eastgate IP) **The patent does not expire until August 11, 2036.** (GameSpot) Monolith Productions (the system's creator) was shut down by WB in February 2025, but the patent remains enforceable. (GameSpot)

For The Concord, this means the Ascendancy cannot use individual NPC memory of specific player encounters that modify NPC progression through a hierarchy — the patent's core claims. However, **tracking collective/aggregate player behavior across populations is legally**

distinct. The Ascendancy should respond to what humanity (all players) does collectively, not what individual encounters produce in individual enemies. This is actually better design for an MMO anyway — the threat should feel like a civilization responding to a civilization, not individual enemies with grudges.

Stellaris shows how to scale existential threats to player power

Stellaris's endgame crisis system offers three distinct threats Gamer Empire — the Prethoryn Scourge (biological invasion, telegraphed ~50 years in advance), Game Rant the Contingency (AI uprising, triggered by synthetic technology research), CBR and the Unbidden (extradimensional portal invasion). Each has a fixed tactical profile: the Scourge uses heavy armor (weak to energy weapons), the Contingency has balanced defenses (no easy counter), and the Unbidden are pure shields (weak to ballistics). This allows **strategic counterplay** — players study the threat and prepare specifically, Steam Community rather than simply accumulating more power.

The **crisis strength multiplier** (configurable from 0.25× to 25.0×) CBR and the "Become the Crisis" ascension path (where players themselves can become the existential threat through a five-level Menace system) TheGamer demonstrate sophisticated scaling. If "All" crises are enabled, each subsequent crisis is **2× more powerful** than the previous. Fandom The crises are telegraphed — the Prethoryn send subspace echoes decades in advance, the Contingency broadcasts a "Ghost Signal" that weakens synthetic populations Game Rant — giving players preparation windows that make the eventual confrontation feel earned.

Dwarf Fortress simulates history through volume, not sophistication

Dwarf Fortress generates up to **2,000 years of pre-game history** Dwarf Fortress Wiki Dwarf Fortress Wiki across maps comparable to the state of Minnesota. It simulates civilizations, wars, trade routes, Gameaipro individual historical figures (with tracked births, deaths, kills, relationships, offspring, and migrations), artifacts, and mythology. Read the Docs Creator Tarn Adams describes the history generator as **"a giant zero-player strategy game with somewhat loose turn rules and bad AI, but thousands of agents."** Dwarf Fortress Wiki Dwarf Fortress Wiki

Adams' four design principles are invaluable: don't overplan (get something running and iterate), break down the system (simulate temperature, rainfall, and drainage separately rather than defining biomes directly), Gameaipro don't overcomplicate (operate at what the player sees or one layer below), and base systems on real-world analogs. gameaipro The key insight is that **emergent complexity comes from volume of simple agents, not sophistication of individual agents**. A thousand orcs with simple AI produce more interesting history than ten orcs with complex AI.

LLMs are best for narrative flavor, not strategic decision-making

As of 2025-2026, **no major shipped AAA title uses LLMs as core AI director systems**. Primary applications are NPC dialogue generation and procedural narrative. NHSJS The academic frontier includes Stanford's "Generative Agents" (2023) — LLM agents simulating daily life with reflection and planning lanbicking — and "The Drama Machine" (2024), which decomposes character simulation into parallel LLM agents with "Ego" and "Superego" roles plus Director and Narrator agents. Emergent Mind GDC 2025-2026 featured talks on "Growing an AI Director into a Full Adventure Director" and "LLM as Core Gameplay," GDC Vault indicating growing industry interest.

The consensus: LLMs excel at **local coherence** (generating believable dialogue, taunts, propaganda, personality expression) but struggle with **macro-level strategic consistency**. For the Ascendancy, this suggests a layered architecture where classical AI handles strategic planning and LLMs handle player-facing communication — making the Ascendancy feel like it has personality without trusting strategic decisions to stochastic language models.

The recommended architecture for the Ascendancy is a four-layer hierarchy

Based on all research, the Ascendancy should use a **hierarchical multi-agent system** with four distinct layers:

- **Strategic Mind** (campaign-level, asynchronous, plans over days/weeks of real time): Uses Hierarchical Task Network planning combined with Monte Carlo Tree Search for long-term strategy across all worlds. "Personality" parameters shape doctrine — aggressive vs. cautious, vengeful vs. calculating, expansionist vs. consolidating. Inputs include all world states and aggregate player behavior patterns. MCTS-HTN hybrids have been validated in RTS environments ScienceDirect and used in Total War: Rome II for campaign-level AI.
- **Sector Commanders** (per-region, utility-based): Multiple agents each managing clusters of worlds, executing the Strategic Mind's plans with local adaptation. Communicate through a Blackboard system (proven in Alien: Isolation). medium Provide the distributed execution layer that prevents single points of failure.
- **Tactical Directors** (per-encounter, real-time): Left 4 Dead-style pacing with stress/relaxation cycles, Alien: Isolation-style menace gauges for tension management, and gated behavior trees that make the Ascendancy appear to "learn" player tactics. Handles spawning, combat behavior, and retreat decisions.
- **Narrative Layer** (LLM-powered): Generates taunts, threats, propaganda, and diplomatic communications. Expresses the Ascendancy's "personality" and references collective player history. Creates emergent story beats without requiring hand-authored content.

Lessons for The Concord's Ascendancy

Telegraph major threats like Stellaris. Ascendancy invasions should be foreshadowed weeks in advance — anomalous Lattice readings, refugee reports from frontier worlds, intelligence fragments. This creates anticipation and allows strategic counterplay.

Use the Director illusion. Name the Ascendancy, give it a communication style, let players attribute intelligence to it. Half of making an AI feel alive is giving players the narrative framework to interpret its actions as intentional.

Track collective behavior, not individual encounters. This avoids the Nemesis patent while being better design for an MMO. The Ascendancy should respond to what weapons are popular across the player base, what defensive strategies dominate, which worlds receive the most investment — and adapt its doctrine accordingly.

Make the personality legible over months. Players should be able to study the Ascendancy's behavior patterns across campaigns and begin to predict its moves. This creates the "chess against a mind three moves ahead" feeling — not through actually being three moves ahead, but by having consistent doctrinal biases that attentive players can learn to read.

Separate omniscience from behavior. The Strategic Mind can know everything, but forces players encounter must operate with realistic information constraints. This creates the feeling of an intelligent adversary without making individual encounters feel unfair.

DOMAIN 4: Long-Arc Phases and Player Lifecycle — Spanning 800 Years

The 4.4% reality and why social bonds trump content

EVE Online's data, presented at EVE North 2019, is sobering: roughly 10,000 new accounts are created per week, **50% of registrants never log in**, and by Day 30, fewer than 500 of each weekly cohort are still playing — a **4.4% 30-day retention rate**. From 2014 CCP data, of those who subscribed, 50% left after their first subscription, 40% ran missions solo until quitting, and **only 10% found a community and stayed**.

Counterintuitively, this is *above* industry average. SuperData Research's five-year study of 1M+ users found post-launch F2P MMOs see only **2-3% 30-day retention**. Players joining 12+ months after launch show just 35% Day 1 return rates, declining to 3% at Day 30. Stabilization occurs roughly 24 months post-launch when word-of-mouth self-qualifying players begin to dominate new arrivals.

The academic consensus is unambiguous: **social interaction is the single strongest predictor of retention across all studies**. A University of Minnesota dissertation and an NCsoft Blade & Soul study (100,000 players) both found that players with more frequent social

interaction churn significantly less. Richard Bartle's player type model, refined by Nick Yee's factor analysis of 7,000 MMO players, suggests that Socializers represent roughly **80% of the player population** — far more than Achievers (~10%), Explorers (~10%), or Killers (~1%). The Concord must design for Socializers first, then layer systems for other types.

Foxhole proves that total reset plus persistent reputation prevents burnout

Foxhole's wars average **20-30 days**, with War 100 stretching to a record-breaking 55.8 days. Between wars, the **Resistance Phase** lasts 1-2 days — everything resets except player rank (commend-based, from other players) and level (activity-based). Players explicitly praise this enforced recovery period for preventing burnout. One community member summarized: "If you aren't exhausted after organizing a regiment, grinding materials, fighting on the front, and actually making a difference for 35 days straight, then you're just a frontline blueberry."

The key design principle: **players accept losing all material progress when social standing persists, shared war stories create permanent social capital, and each cycle feels genuinely different**. New players face no equipment barriers — they can operate any vehicle or weapon from Day 1, and rank is purely cosmetic/social prestige. This creates the conditions for Foxhole's most remarkable emergent property: logistics players (scrapping, manufacturing, transport) are valued as highly as combat players, because the game's interdependence makes both essential.

No Man's Sky proves that Phase transitions can function as re-launch events

No Man's Sky's player count trajectory is the most dramatic recovery story in gaming history. After launching to a **212,613 concurrent player peak** in August 2016 and collapsing to ~1,000-2,000 average players by late 2016, Hello Games executed a multi-year resurrection through free content updates. The NEXT update (July 2018) hit 97,723 concurrent. Beyond (August 2019) reached 61,535. Most remarkably, the Voyagers update in August 2025 hit **110,893 concurrent players** — the highest since launch and nearly matching it nine years later.

The pattern is consistent: major feature updates create **10-50× spikes** from baseline, which decay but leave the baseline higher each time. All updates are completely free, each targets a specific player desire (multiplayer, VR, building, combat), and cumulative improvement means each return feels dramatically different. **Each Phase of The Concord can function as a re-launch event** following this model — but only if each Phase feels transformatively different from the last, entry barriers for returning players are minimal, and the changes are dramatic enough to generate press coverage and word-of-mouth.

FFXIV's End of an Era is the gold standard for Phase transitions

The most dramatic Phase transition in MMO history was Final Fantasy XIV's transition from 1.0 to 2.0. Rather than quietly sunsetting a critically failed game, Square Enix built the "Seventh Umbral Era" storyline: the moon Dalamud fell, revealing the imprisoned primal Bahamut, who

devastated the world of Eorzea. **Players who were logged in during the final server shutdown witnessed the destruction in real time** — an unforgettable, unrepeatable experience. FFXIV 2.0 (A Realm Reborn, 2013) launched as a "new game" set in the aftermath and became one of the most successful MMOs ever.

Legacy players received special tattoo markings and were acknowledged in-game as "Warriors of Light" who survived the Calamity. The key lessons: catastrophic transitions generate more engagement than gradual changes, **witnessing the transition creates permanent social prestige**, and framing destruction as narrative (not failure) transforms a technical reset into a cultural event.

Guild Wars 2's Living World Season 1 offers a cautionary counterpoint: Scarlet Briar's storyline permanently destroyed Lion's Arch (the game's main hub), but the content was ephemeral — events happened once and were gone. Players who missed Season 1 had a massive narrative gap. ArenaNet spent 2022 rebuilding the content as permanent, replayable episodes. **The Concord must preserve Phase history** so that new players can experience or at least learn what happened, not be excluded by it.

The bitter vet problem and why evergreen systems matter more than content

EVE's "bitter vet" phenomenon — long-time players who have exhausted content, become cynical about changes, and are vocally negative yet continue subscribing — is structural, not behavioral. Veterans accumulate resources faster than content can consume them, creating "solved game" syndrome. CCP's responses have included skill extraction/injection (2016, allowing character restructuring), Alpha clones (F2P tier), seasonal events, and the CSM as a voice for veterans. None fully solve the problem because **the core issue is that vertical progression has a ceiling while horizontal options expand indefinitely**.

For a game spanning 5-10+ real years across five Phases, this means each Phase cannot simply add more levels or more powerful gear. The Concord needs **evergreen systems that scale with player experience** — governance roles that become more complex, mentor relationships that reward teaching, historical legacy systems that give veterans unique narrative positions, and new frontiers (literally, as the Lattice expands from a handful of worlds to 600+) that provide fresh challenges without invalidating veteran expertise.

The new player in an old world needs asymmetric value, not catch-up mechanics

The "new player in an old world" problem has been approached in three fundamentally different ways. WoW uses **gear catch-up mechanics** — each patch makes previous gear irrelevant, letting returners reach near-current power quickly but devaluing veteran investment. FFXIV requires **the full journey** — all players must play through the entire Main Scenario Quest from the beginning, creating shared cultural references but requiring enormous time investment (Square Enix trimmed the notoriously tedious post-ARR content in patch 5.3). EVE uses

asymmetric specialization — new players can never match veterans in total skill points but can be combat-effective in specific roles (tackle frigates in fleet fights) within days.

For The Concord, EVE's asymmetric approach is most appropriate. A Phase 4 newcomer should be immediately useful in specific roles while veterans provide breadth and leadership. Foxhole's "no equipment barriers" principle applies: any new player should be able to operate meaningful systems from Day 1, with expertise providing efficiency and options rather than access. In-game archives, museums, and lore repositories should make the world's history a selling point rather than an exclusion mechanism. Mentorship systems should give veterans tangible KALON rewards for training newcomers, creating an economic incentive for inclusion.

Lessons for The Concord's five-Phase arc

Make Phase transitions participatory events, not content patches. Follow FFXIV's End of an Era model: the transition itself should be playable content that players witness, with those present receiving permanent recognition. Each Phase transition should generate media coverage and function as a re-launch (No Man's Sky model).

Preserve history, don't erase it. Unlike GW2's ephemeral Season 1, build permanent in-game archives of Phase history. New players entering in Phase 4 should feel excited by five years of player-created history, not excluded by it. Consider "historical instance" content that lets new players experience key moments from earlier Phases.

Design multiple engagement cadences. Short-term daily activities (alerts, Ascendancy skirmishes), medium-term weekly campaigns, long-term Phase-spanning narratives, and ultra-long-term cross-Phase legacy systems. Not every player needs to engage at every cadence — the system should accommodate players who engage intensely for a month, leave for six months, and return for the next Phase transition.

Accept and design for the 4.4% reality. Invest maximally in making social connection happen in the first week. The 10% of players who find community stay for years; the other 90% will churn regardless of content quality. Every system should be designed to create reasons for players to interact, depend on each other, and form bonds — from Foxhole-style logistics interdependence to governance participation to Ascendancy defense coordination.

Use the expanding Lattice as the solution to the bitter vet problem. Starting with a handful of connected worlds and expanding to 600+ over years provides an ever-expanding frontier. Veterans who have "solved" the core worlds become the explorers, governors, and defenders of new Lattice nodes. Each expansion is both new content and new social territory — new worlds need new governors, new economies, new defensive networks.

Reward witnesses and pioneers at every Phase. Players who were present for Phase 1 should carry permanent, visible markers of their legacy — titles, cosmetics, Assembly privileges, historical recognition. But these markers should convey prestige, not power. A Phase 4 newcomer should be able to defeat a Phase 1 veteran in combat or outperform them

economically; the veteran's advantage is social capital and institutional knowledge, not mechanical superiority.

Conclusion: The Concord as integrated system

The most important finding across all four domains is that **these systems cannot be designed in isolation**. The KALON economy determines what governance means (because voting weight could be tied to contribution). Governance determines how players collectively respond to the Ascendancy (because the Assembly sets defensive priorities). The Ascendancy determines what economic activity matters (because worlds under threat need different contributions than peaceful ones). And the Phase arc determines when each system matters most (because Phase 1's handful of worlds creates fundamentally different economic and political dynamics than Phase 5's 600+ worlds).

Three cross-cutting principles emerge. First, **social bonds are the load-bearing structure** — economy, governance, AI threat, and progression are all mechanisms for creating, testing, and strengthening relationships between players. Second, **transparency builds trust** — publish economic data like CCP's MERs, make governance votes public, explain the Ascendancy's behavioral patterns to players who study them, and treat the community as partners rather than consumers. Third, **destruction creates meaning** — EVE's economic cycles, Foxhole's war resets, FFXIV's Calamity, the Ascendancy consuming worlds — loss is what makes investment feel real.

The Concord's most radical design proposition — that value is measured in contribution, not accumulation — has never been tested at MMO scale. But the research suggests it aligns with how persistent virtual worlds actually work. Castronova found that even without subsistence needs, players organize around social contribution. EVE's most retained players are those who found a community to contribute to. Foxhole's logistics players are valued not for what they accumulate but for what they deliver. The KALON system does not fight human nature — it codifies what already makes virtual civilizations work.